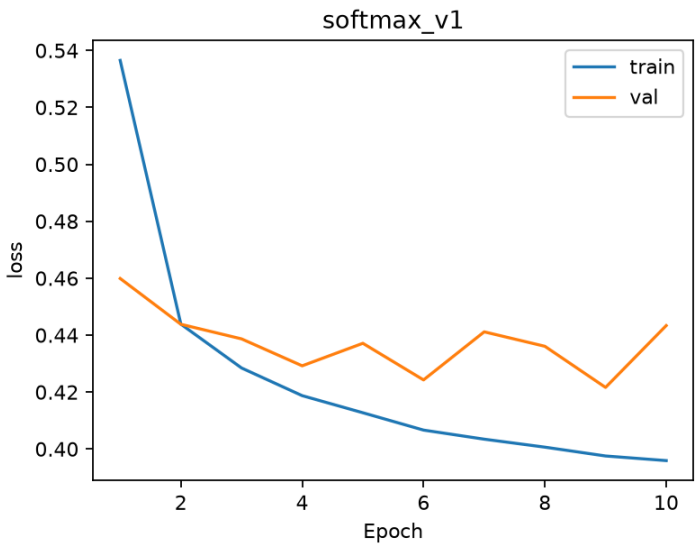
E1 - Fashion-MNIST

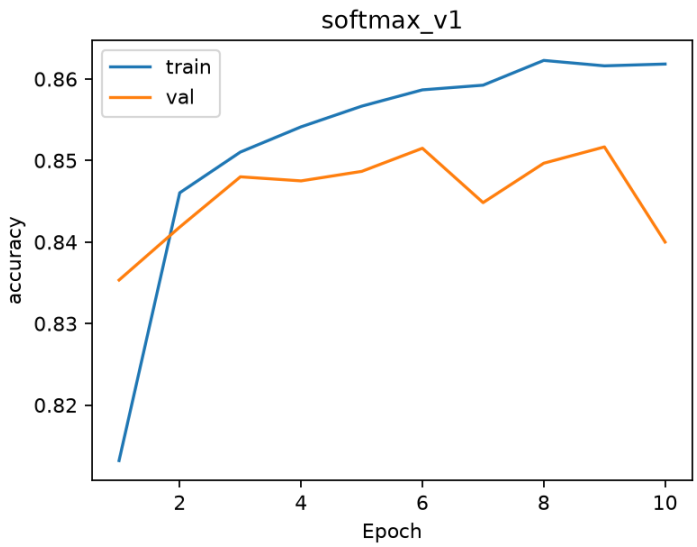
Bùi Tiến Đức - 2680018. 54,000 train / 6,000 validation / 10,000 test. Seed 42; 10 epochs. Checkpoint selected by validation accuracy, earliest tie.

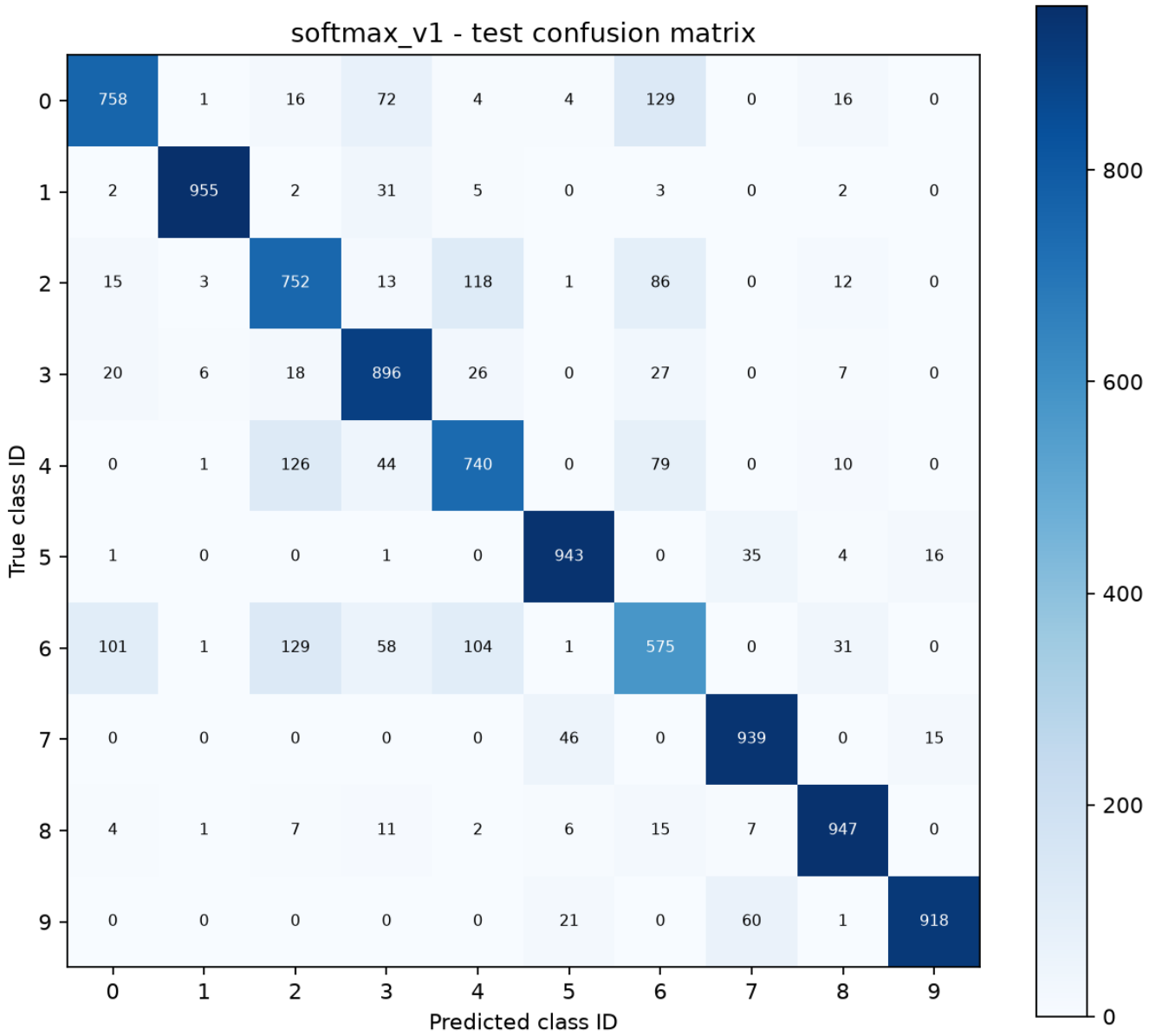
Run	Best epoch	Parameters	Val accuracy	Test accuracy	Test Macro-F1
softmax_v1	9	7850	85.17%	84.23%	84.14%
mlp_v1	9	235146	89.17%	88.40%	88.31%
cnn_v1	8	206922	91.63%	91.28%	91.26%
mlp_dropout05_v1	9	235146	87.93%	87.31%	87.14%

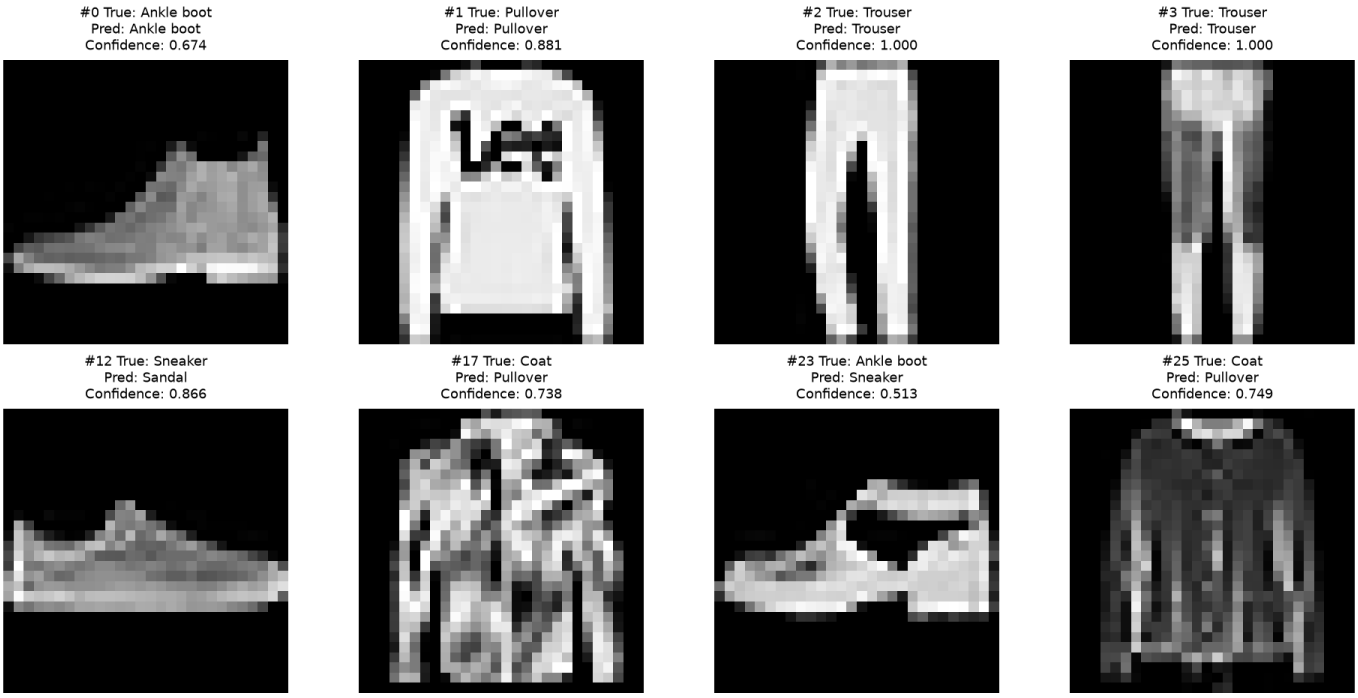
One seed only. Dropout experiment is a limited adaptation, not a complete reproduction of the original paper. Training accuracy uses train mode; validation uses eval mode. Timing includes train and validation loops, not full wall time. Test outcomes must not be used for further tuning.

softmax_v1

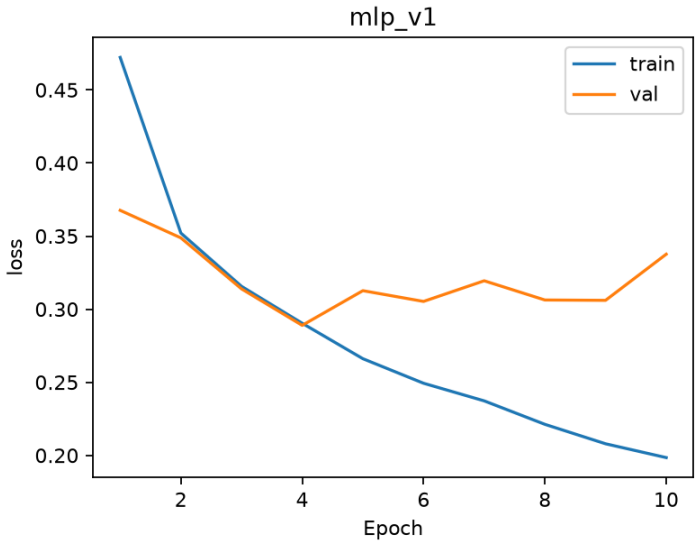


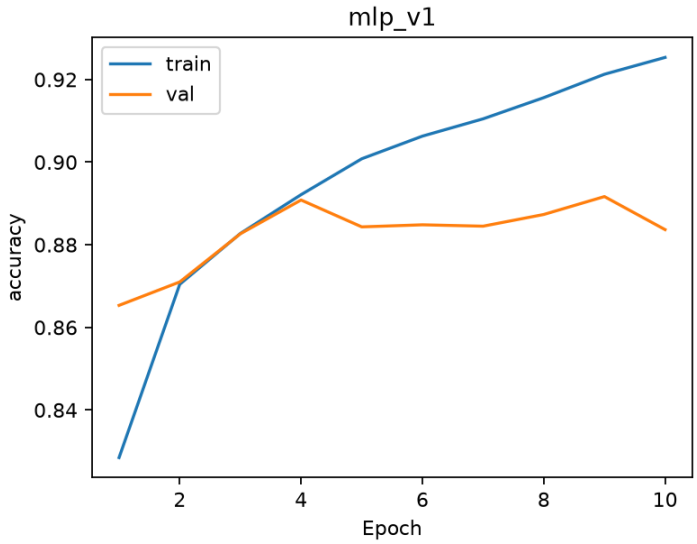


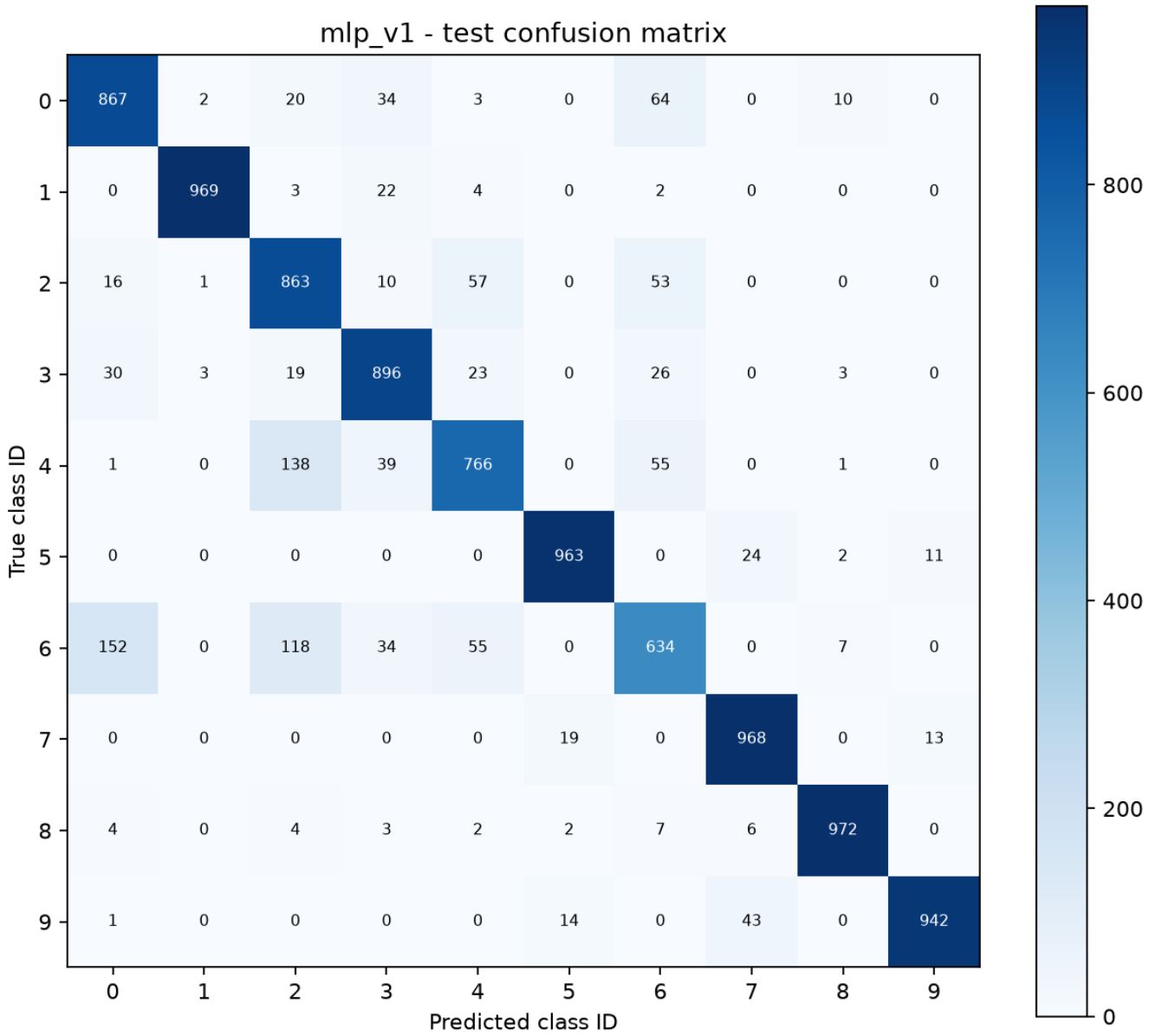

softmax_v1 - test confusion matrix
True class ID
Predicted class ID
0 1 2 3 4 5 6 7 8 9
758 1 16 72 4 4 129 0 16 0
2 955 2 31 5 0 3 0 2 0
15 3 752 13 118 1 86 0 12 0
20 6 18 896 26 0 27 0 7 0
0 1 126 44 740 0 79 0 10 0
1 0 0 1 0 943 0 35 4 16
101 1 129 58 104 1 575 0 31 0
0 0 0 0 0 46 0 939 0 15
4 1 7 11 2 6 15 7 947 0
0 0 0 0 0 21 0 60 1 918
800
600
400
200
0

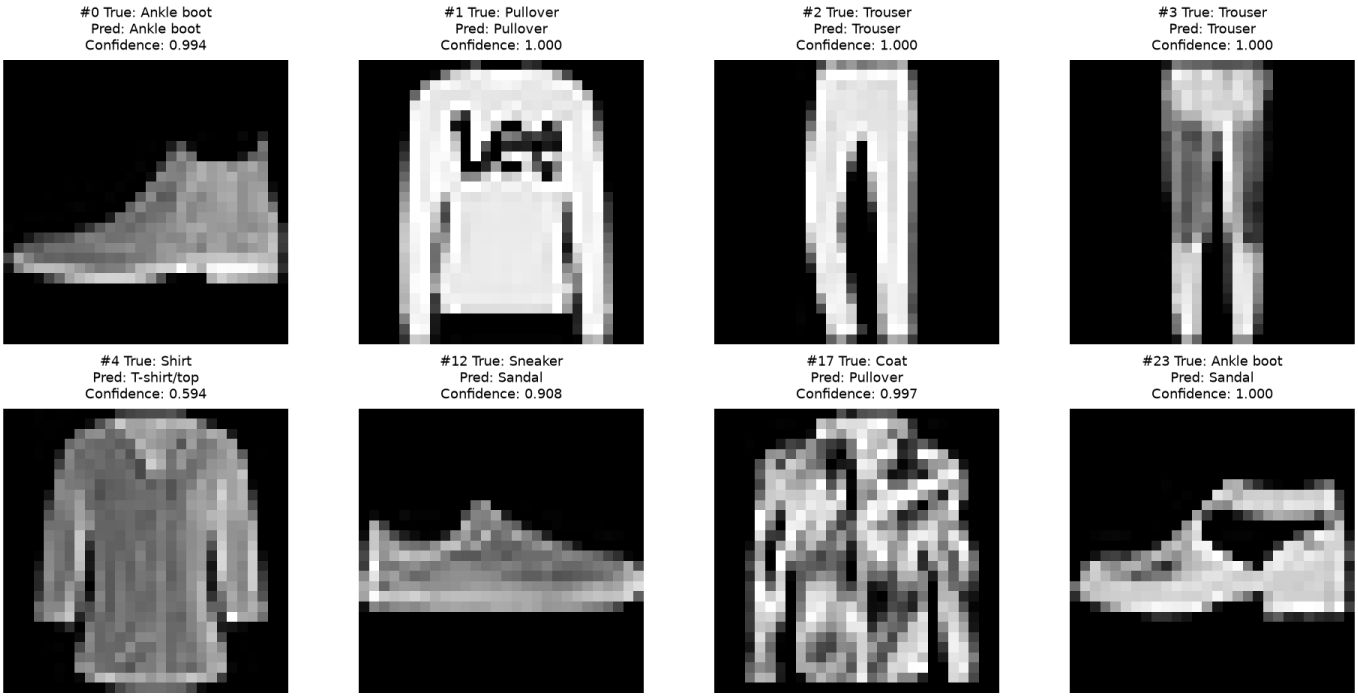

#0 True: Ankle boot
Pred: Ankle boot
Confidence: 0.674
#1 True: Pullover
Pred: Pullover
Confidence: 0.881
#2 True: Trouser
Pred: Trouser
Confidence: 1.000
#3 True: Trouser
Pred: Trouser
Confidence: 1.000
#12 True: Sneaker
Pred: Sandal
Confidence: 0.866
#17 True: Coat
Pred: Pullover
Confidence: 0.738
#23 True: Ankle boot
Pred: Sneaker
Confidence: 0.513
#25 True: Coat
Pred: Pullover
Confidence: 0.749

mlp_v1

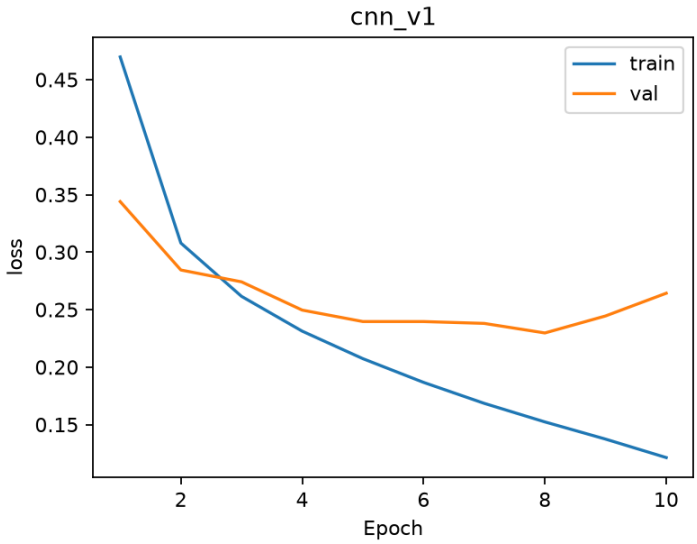

mlp_v1
train
val
loss
Epoch
0.45
0.40
0.35
0.30
0.25
0.20
2
4
6
8
10

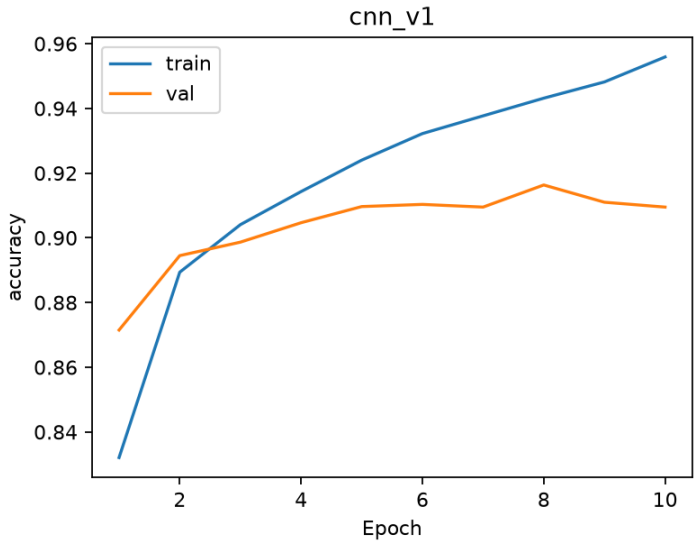

mlp_v1
train
val
accuracy
Epoch
0.92
0.90
0.88
0.86
0.84
2
4
6
8
10

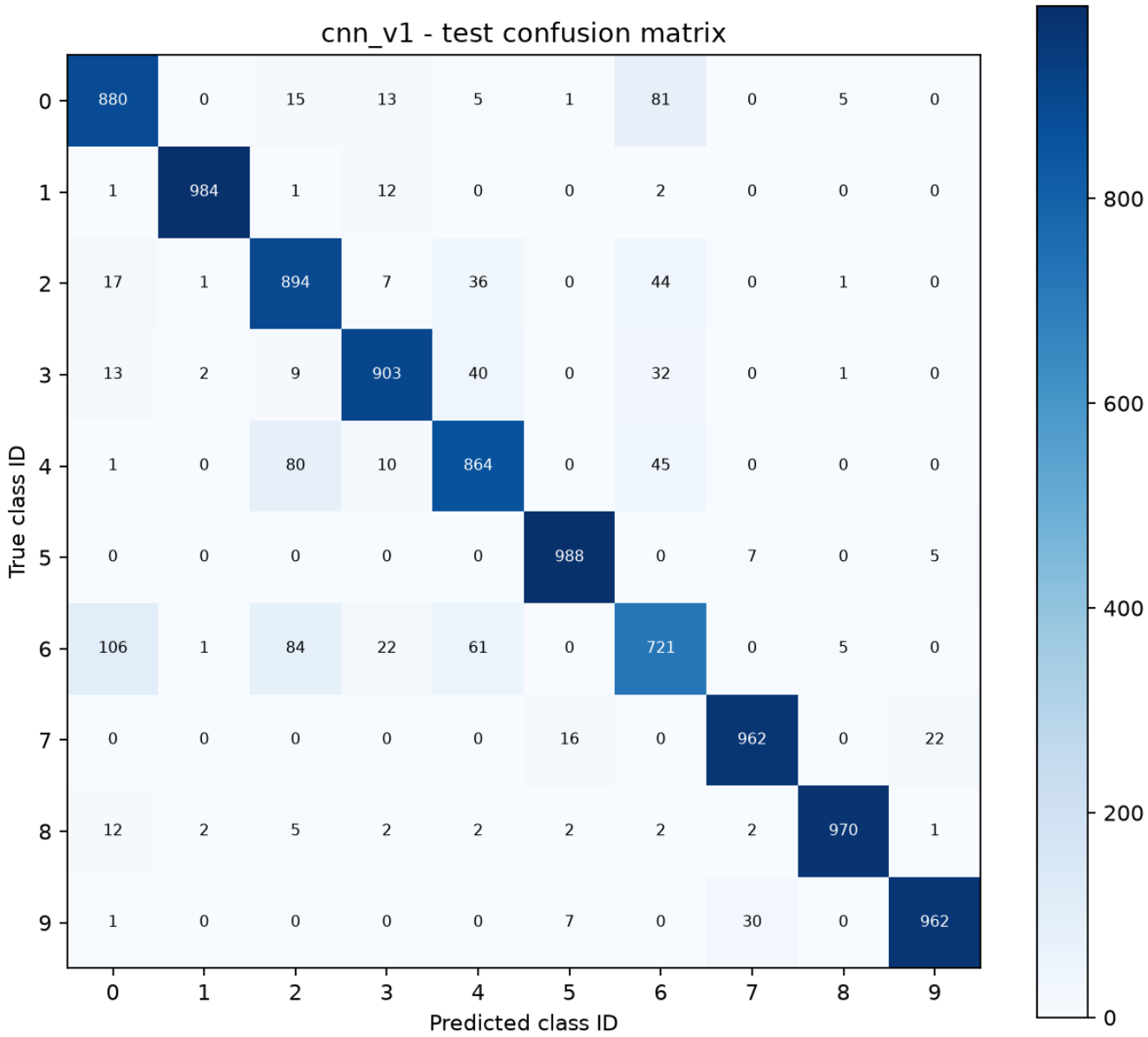

mlp_v1 - test confusion matrix
True class ID
Predicted class ID
0 867 2 20 34 3 0 64 0 10 0
1 0 969 3 22 4 0 2 0 0 0
2 16 1 863 10 57 0 53 0 0 0
3 30 3 19 896 23 0 26 0 3 0
4 1 0 138 39 766 0 55 0 1 0
5 0 0 0 0 0 963 0 24 2 11
6 152 0 118 34 55 0 634 0 7 0
7 0 0 0 0 0 19 0 968 0 13
8 4 0 4 3 2 2 7 6 972 0
9 1 0 0 0 0 14 0 43 0 942
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

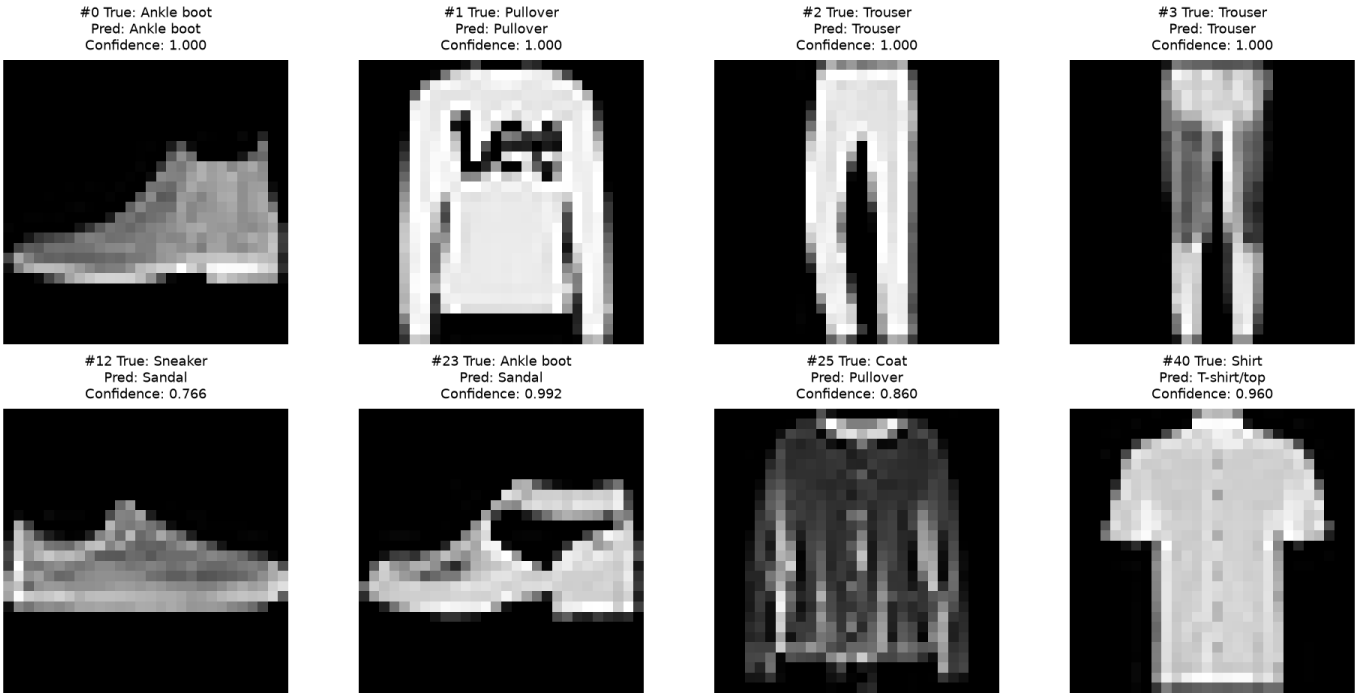

#0 True: Ankle boot
Pred: Ankle boot
Confidence: 0.994
#1 True: Pullover
Pred: Pullover
Confidence: 1.000
#2 True: Trouser
Pred: Trouser
Confidence: 1.000
#3 True: Trouser
Pred: Trouser
Confidence: 1.000
#4 True: Shirt
Pred: T-shirt/top
Confidence: 0.594
#12 True: Sneaker
Pred: Sandal
Confidence: 0.908
#17 True: Coat
Pred: Pullover
Confidence: 0.997
#23 True: Ankle boot
Pred: Sandal
Confidence: 1.000

cnn_v1

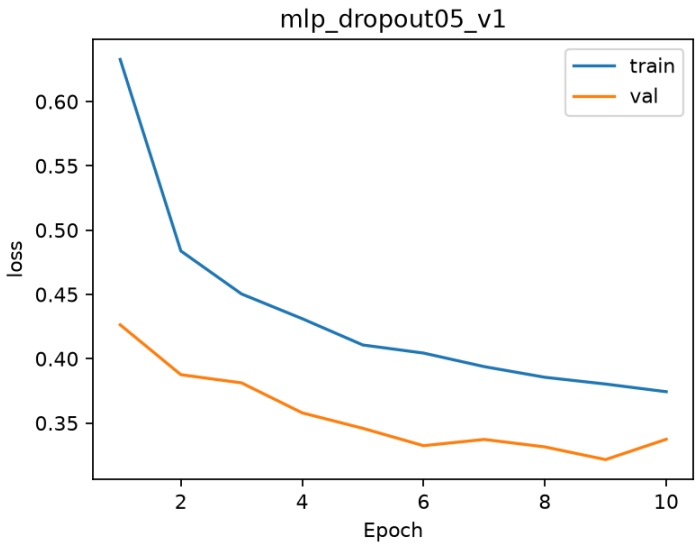

cnn_v1
train
val
loss
Epoch
0.45
0.40
0.35
0.30
0.25
0.20
0.15
2
4
6
8
10

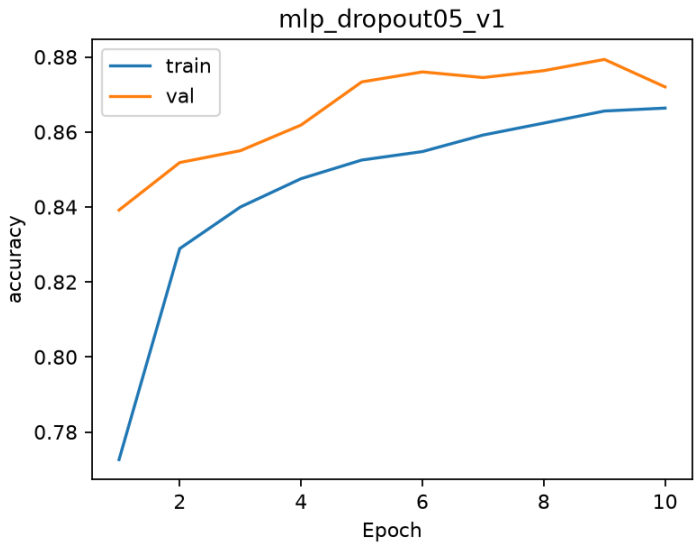

cnn_v1
train
val
accuracy
Epoch
0.96
0.94
0.92
0.90
0.88
0.86
0.84
2
4
6
8
10

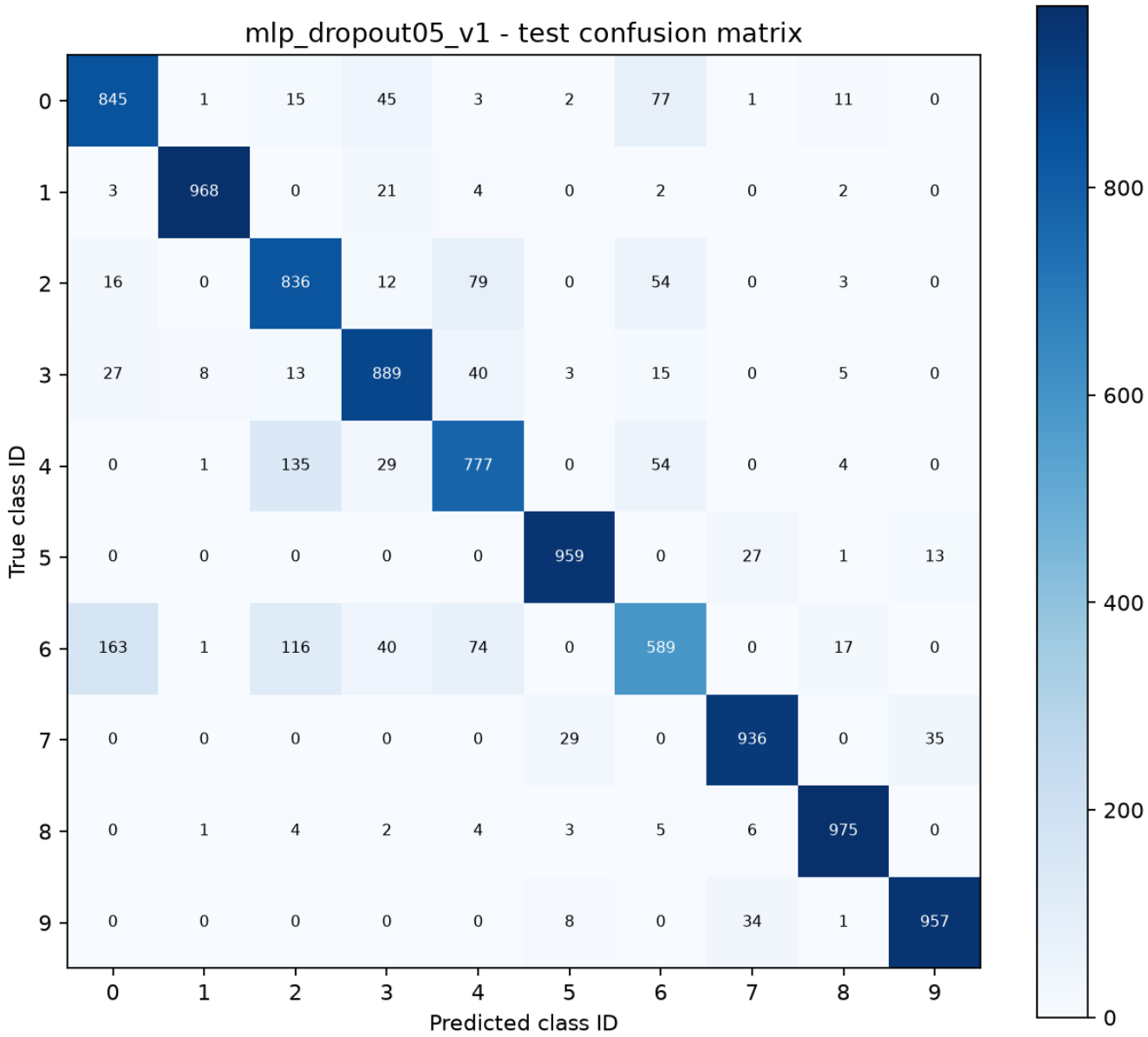

cnn_v1 - test confusion matrix
True class ID
Predicted class ID
0 880 0 15 13 5 1 81 0 5 0
1 1 984 1 12 0 0 2 0 0 0
2 17 1 894 7 36 0 44 0 1 0
3 13 2 9 903 40 0 32 0 1 0
4 1 0 80 10 864 0 45 0 0 0
5 0 0 0 0 0 988 0 7 0 5
6 106 1 84 22 61 0 721 0 5 0
7 0 0 0 0 0 16 0 962 0 22
8 12 2 5 2 2 2 2 2 970 1
9 1 0 0 0 0 7 0 30 0 962
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

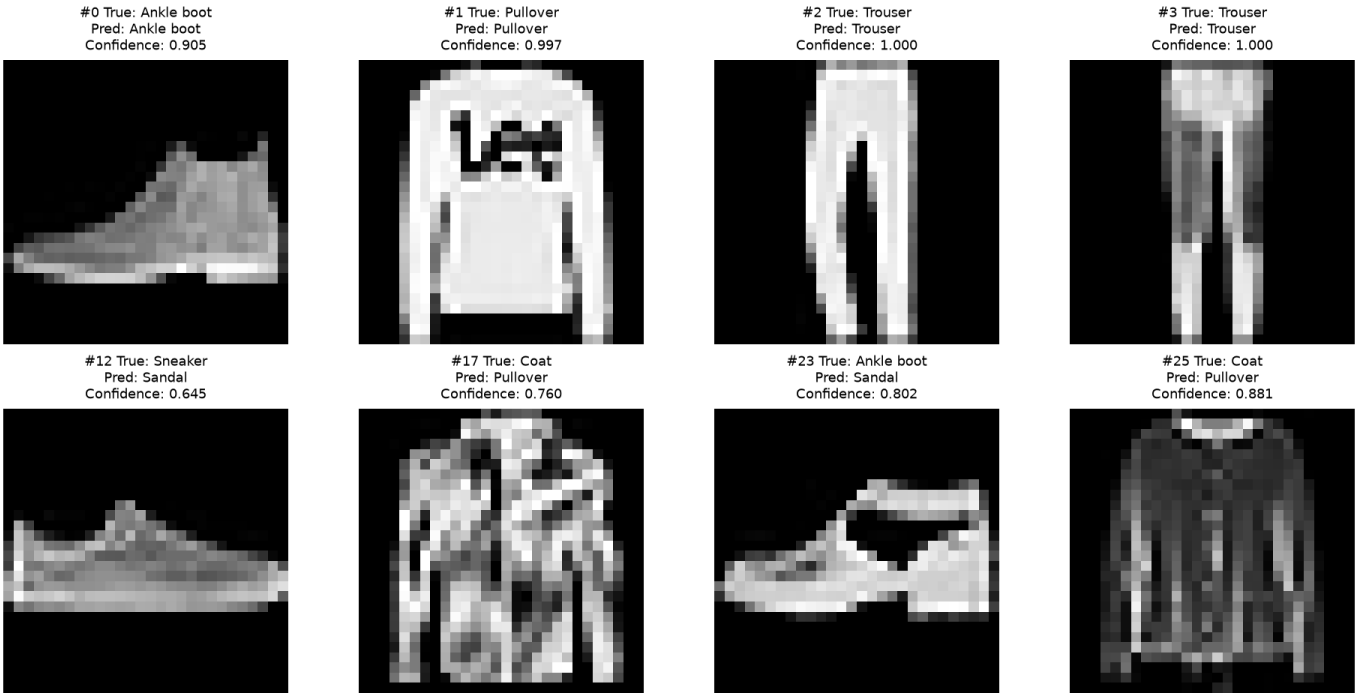


mlp_dropout05_v1






mlp_dropout05_v1 - test confusion matrix
True class ID
Predicted class ID
0 1 2 3 4 5 6 7 8 9
845 1 15 45 3 2 77 1 11 0
3 968 0 21 4 0 2 0 2 0
16 0 836 12 79 0 54 0 3 0
27 8 13 889 40 3 15 0 5 0
0 1 135 29 777 0 54 0 4 0
0 0 0 0 0 959 0 27 1 13
163 1 116 40 74 0 589 0 17 0
0 0 0 0 0 29 0 936 0 35
0 1 4 2 4 3 5 6 975 0
0 0 0 0 0 8 0 34 1 957
800
600
400
200
0


#0 True: Ankle boot
Pred: Ankle boot
Confidence: 0.905
#1 True: Pullover
Pred: Pullover
Confidence: 0.997
#2 True: Trouser
Pred: Trouser
Confidence: 1.000
#3 True: Trouser
Pred: Trouser
Confidence: 1.000
#12 True: Sneaker
Pred: Sandal
Confidence: 0.645
#17 True: Coat
Pred: Pullover
Confidence: 0.760
#23 True: Ankle boot
Pred: Sandal
Confidence: 0.802
#25 True: Coat
Pred: Pullover
Confidence: 0.881